

NAME

AHMED IKRAM RAO

REG. NO

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(ASSIGNMENT-2)

QUESTION #01

i) Find the transition probabilities:-

→ Transitions from state 0 :-

$$0 \rightarrow 0 = 8$$

$$0 \rightarrow 1 = 12$$

$$\text{Total} = 32$$

$$0 \rightarrow 2 = 12$$

→ Transitions from state 1

$$1 \rightarrow 0 = 5$$

$$\text{Total} = 23$$

$$1 \rightarrow 1 = 2$$

$$1 \rightarrow 2 = 16$$

→ Transitions from state 2

$$2 \rightarrow 0 = 2$$

$$2 \rightarrow 1 = 2$$

$$\text{Total} = 4$$

$$2 \rightarrow 2 = 0$$

$$\text{Total Transitions} = 59$$

Now probabilities

State 0

$$P_{00} = 8/32 = 0.25$$

$$P_{01} = 12/32 = 0.375$$

$$P_{02} = 12/32 = 0.375$$



state 1

$$P_{10} = 5/23 = 0.2173$$

$$P_{11} = 2/23 = 0.0869$$

$$P_{12} = 16/23 = 0.6956$$

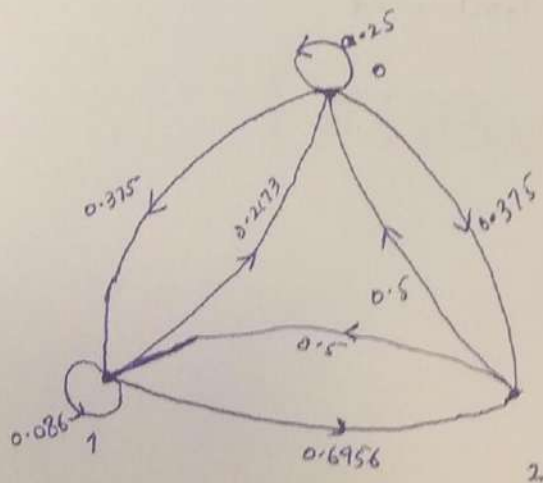
state 2

$$P_{20} = 2/4 = 0.5$$

$$P_{21} = 2/4 = 0.5$$

$$P_{22} = 0/4 = 0$$

⑥ state transition diagram (weighted directed graph)



d) Transition probability matrix

$$P = \begin{bmatrix} 0.25 & 0.375 & 0.375 \\ 0.2173 & 0.0869 & 0.6956 \\ 0.5 & 0.5 & 0 \end{bmatrix}$$

c) what is mean P_{01}^4 ?

$$P^4 = P \times P \times P \times P$$

$$P^4 = \begin{bmatrix} 0.407 & 0.323 & 0.270 \\ 0.392 & 0.316 & 0.292 \\ 0.431 & 0.332 & 0.237 \end{bmatrix}$$

Now

P_{01}^4 will be 0.323

b) It's a Markov chain with 3 states

$$S = \{0, 1, 2\}$$

Transition probabilities given by matrix P_2

$$P_2 = \begin{bmatrix} P_{00} & P_{01} & P_{02} \\ P_{10} & P_{11} & P_{12} \\ P_{20} & P_{21} & P_{22} \end{bmatrix}$$

$$P_2 = \begin{bmatrix} 8/32 & 12/32 & 12/32 \\ 5/23 & 2/23 & 16/23 \\ 2/4 & 2/4 & 0/4 \end{bmatrix}$$